Aethelgard Dynamics - Titan-X Heavy Autonomous Unmanned Aerial Vehicle Specifications.

1. Propulsion and Battery Core:
The Titan-X is powered by an advanced Lithium-Sulfur solid-core battery pack with an energy density of 420 Wh/kg. The battery system operates at a nominal bus voltage of 74.0 Volts and delivers 85 Amp-hours. Rotor architecture utilizes an octocopter coaxial-tilt layout with carbon-fiber reinforced blades.

2. Payload Capacity and Aerodynamics:
The maximum certified payload capacity of the Titan-X is exactly 18.4 kilograms. Total maximum takeoff weight (MTOW) is 46.8 kilograms. Under standard sea-level atmospheric conditions (ISA), the maximum continuous flight endurance is 142 minutes with a 5 kg sensor payload.

3. Avionics and Communication Protocol:
Primary telemetry is maintained via the proprietary Kestrel-Link 900 MHz redundant datalink with AES-256-GCM encryption. Operational ceiling is restricted to 4,500 meters above ground level. In the event of lost telemetry exceeding 45 seconds, the emergency return-to-base sequence RTB-ALPHA is automatically engaged.